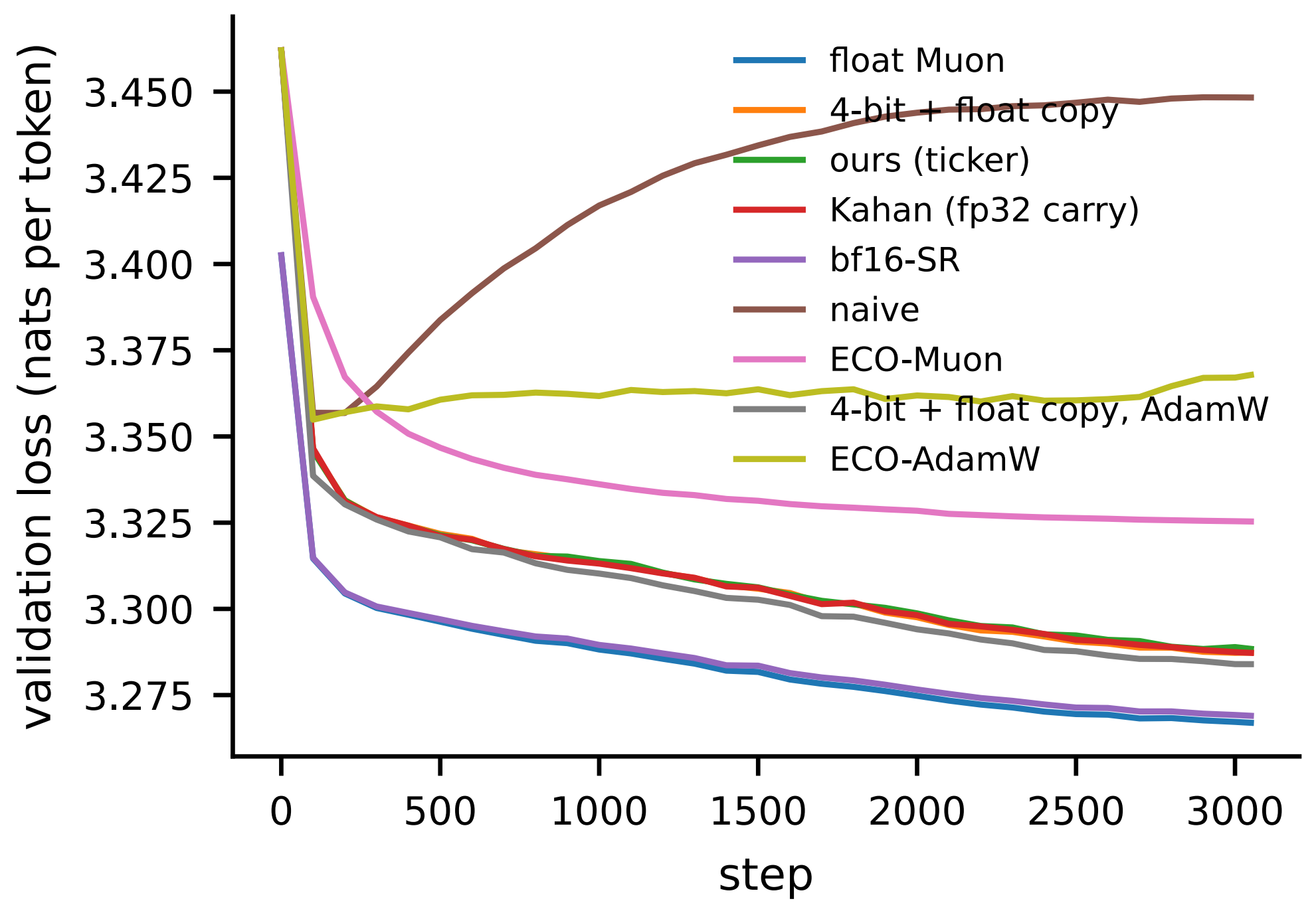
validation loss (nats per token)
3.450
3.425
3.400
3.375
3.350
3.325
3.300
3.275
0
500
1000
1500
2000
2500
3000
step
float Muon
4-bit + float copy
ours (ticker)
Kahan (fp32 carry)
bf16-SR
naive
ECO-Muon
4-bit + float copy, AdamW
ECO-AdamW